Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] **[18EE61]**

SIXTH SEMESTER B.Tech EXAMINATION, JULY – 2022

POWER SYSTEM -II

Time:3 Hours **Maximum Marks:70**

Instructions:

i. Answer any **FIVE** full Questions. Questions 1 and 2 are compulsory. Each question carries 14 marks.

ii. Missing data may be suitably assumed.

1. a. Show that per unit impedance of a transformer remains same whether it is referred H.V. or L.V. side **04**

b. Derive an expression for 3-ph power in terms of symmetrical components **04**

c. A 15 MVA, 8.5 kV, 3-ph generator has a reactance of 20%. It is connected through Delta-star Transformer to a transmission line having a total series reactance of 70 ohms. At the load end of the line is star-star step down transformer. Both the transformer banks are composed of three, single phase transformers connected for 3-ph operation. Each 1-ph units are rated 6.667 MVA, 10kV / 100 kV with reactance of 10%. The load is represented by an impedance drawing 10MVA at 12.5 kV, 0.8 pf lag. Draw the reactance diagram on a base of 10 MVA, 12.5 kV in the load circuit. **06**

2. a. For the power system network shown in fig-2, the ratings of various components are **08**

Generator G : 100 MVA, 11 kV, X″ = 0.2 pu

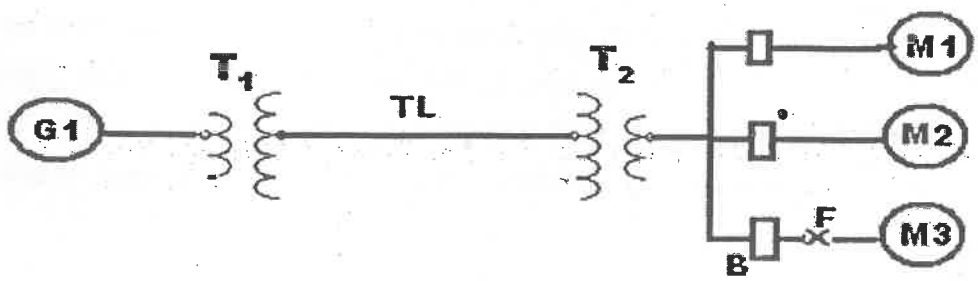
Motor M1, M2 and M3 are of same rating : 20 MVA, 33 kV, X″ = 0.2 pu, X′ = 0.25 pu Transformer T1: 100 MVA, 11 /66 kV, X=10% , T2 : 100 MVA, 66/33 kV, X=10%. Transmission line impedance is 30 ohms, Choose a base of 100 MVA, 11 kV in generator circuit. When a three phase balanced fault occurs at the point F. calculate

i. Subtransient current in the fault

ii. Subtransient current in the circuit breaker B.

iii. Momentary current in the circuit breaker B

iv. The current to be interrupted by circuit breaker B in 2 cycles



b. Two 11 kV, 20 MVA, 3-ph star connected Generators with one generator neutral grounded through 2 ohm resistor and another generator neutral ungrounded, operate in parallel respectively. The positive, negative and zero sequence reactances of each being j0.18, j0.15 and j0.1 pu respectively. A line to ground fault occurs at the terminals of one of the generator. Estimate **06**

i. Fault current

ii. Current flowing through grounding resistor

iii. Voltage across grounding resistor

3. a. Compare the load flow methods **06**

b. Write the algorithm for Newton Raphson method **08**

OR

4. a. Draw the flow chart for Gauss-Seidal iterative solution of load flow **06**

b. Obtain the Gauss-seidal load flow solution at the end of first iteration of the power system. The data is given in Table 4a and 4b. obtain the solution when all buses except bus-1 are PQ buses. Take acceleration factor as 1.6 **08**

Tabale 4a Line data

Bus code	Admittance
1-2	2-j8
1-3	1-j4
2-3	0.666-j2.664
2-4	1-j4
3-4	2-j8

Table.4b Bus data

Bus no	P_i (pu)	Q_i (pu)	V_i
1	---	----	$1.06\angle 0°$
2	0.5	0.2	---
3	0.4	0.3	---
4	0.3	0.1	---

5. a. Explain the thermal constraints in Unit commitment problem **04**

b. Explain the following terms with reference to thermal plants **03**

i. heat rate curve

ii. incremental fuel rate curve

iii. incremental production cost curve

c. Two generators of 200 MW at Bus-1 and 75 MW at Bus-2 are connected through a transmission line. The load is connected at Bus-1. It is observed that when a power of 75MW is imported to bus 1, the loss amounted to 5MW. Find the generation needed from each plant and also the power received by the load, if the system λ is given by Rs 20/MWHr. The incremental fuel cost at the two plants are given by **07**

$$\frac{dC_1}{dP_1} = 15 + 0.03\,P_1$$

$$\frac{dC_2}{dP_2} = 18 + 0.05\,P_2$$

OR

6. a. Derive the expression for Transmission loss B-Coefficients **07**

b. A power system with two generating stations supplied a total load of300MW. Neglecting transmission losses, the economic schedule for the plant generation is 175MW and 125MW. Find the saving in the production cost in Rs/hr. due to this economic schedule as compared to equal distribution of the same load between the two units. The incremental cost characteristics are **07**

$$\frac{dC_1}{dP_1} = 30 + 0.3\,P_1$$

$$\frac{dC_2}{dP_2} = 32.5 + 0.4\,P_2$$

7. a. Derive the power angle equation of a non-salient pole synchronous machine and draw the power angle curve **07**

b. Find the steady state stability limit of a system consisting of a generator of equivalent reactance 0.5pu connected to an infinite bus through a series reactance of 1pu. The terminal voltage of the generator is held at 1.2pu and voltage of the infinite bus is 1 pu. **07**

OR

8. a. With a neat sketch explain equal area criteria of stability for a sudden change in the mechanical input **07**

b. A generator having H= 6 MJ/MVA is delivering a power of 1 pu to an infinite bus through a purely reactance network, when a fault occurred the generator power reduces to zero. The maximum power that could be delivered by the generator is 2.5pu. When the fault is cleared, the original network conditions again exist. Determine the critical clearing angle and critical clearing time **07**